Huawei Access Node Optical Transport Failure Resolution SOP (HEB-ACC-75)

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7579

Description

The Hebbal Edge Huawei access node (HEB-ACC-75) Optical Transport Failure (OTF) is primarily attributed to the degradation of SFP/XFP transceivers, fiber cuts, or a reduction in receive/transmit power. This failure domain occurs on the Access role due to the reliance on optical transport mechanisms for signal transmission and reception. The OTF can be further exacerbated by the inherent limitations of SFP/XFP transceivers, such as temperature sensitivity, optical fiber attenuation, and optical signal degradation due to aging. The Huawei NE4000E access node's optical transport failure is typically indicated by the ALARM_7579 alarm code with a severity of minor. However, it is crucial to promptly address and resolve this issue to prevent potential service outages and ensure optimal network performance. This SOP outlines the procedures for diagnosing and resolving OTF on the HEB-ACC-75 Huawei access node.

Pre-requisite Checks

- pre-requisite: Verify the site location and access node ID to ensure accuracy before initiating troubleshooting.

- pre-requisite: Check the redundant link status to ensure that it is operational and can take over in case of a failure.

- pre-requisite: Dispatch remote hands to the Hebbal Edge site for physical access and assistance.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers	High
Fiber cuts	Medium
Reduction in receive/transmit power	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical transceiver status, Expected Output: Transceiver status: Normal, Description: This command displays the status of the optical transceivers to determine if they are functioning correctly.

Command: display fiber optic cable status, Expected Output: Fiber optic cable status: Normal, Description: This command displays the status of the fiber optic cables to determine if there are any issues with the cables themselves.
```

```
Command: display optical signal quality, Expected Output: Optical signal quality: Good, Description: This command displays the quality of the optical signal to determine if it is within the acceptable range.
```

Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP Transceivers, Command: replace optical transceiver, Description: Replace the degraded SFP/XFP transceivers with new ones to restore normal operation.

Step 2: Step: Repair or Replace Fiber Optic Cables, Command: repair fiber optic cable, Description: Repair or replace the fiber optic cables to restore normal operation.

Step 3: Step: Adjust Optical Signal Quality, Command: adjust optical signal quality, Description: Adjust the optical signal quality to ensure it is within the acceptable range.

Post-Check Verification

```
Command: display optical transceiver status, Expected Output: Transceiver status: Normal, Description: Verify that the optical transceivers are functioning correctly after the resolution steps.

Command: display fiber optic cable status, Expected Output: Fiber optic cable status: Normal, Description: Verify that the fiber optic cables are functioning correctly after the resolution steps.

Command: display optical signal quality, Expected Output: Optical signal quality: Good, Description: Verify that the optical signal quality is within the acceptable range after the resolution steps.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: undo replace optical transceiver, Description: Restore the original configuration by undoing the replace optical transceiver command.

- Step: Restore Original Fiber Optic Cables, Command: undo repair fiber optic cable, Description: Restore the original fiber optic cables by undoing the repair fiber optic cable command.